StatsMeth Exam 2026/7 -

Remark: Notation

- Typically X for discrete RV, Y for continuous

Probability/Statistics Refresh

Recall: Basics

- **pmf** - discrete - $\mathbb{P}[X=x]$ - $f_X(x)$. $0 \le f_X(x) \le 1$, $\sum_{x\in\Omega} f_X(x) = 1$.
- **pdf** - continuous - $\mathbb{P}[a \le Y \le b] = \int_a^b f_Y(y)\ \mathrm{dy}$. $\mathbb{P}[Y=y]=0$, $f_Y(y) \ge 0$, $\int_{-\infty}^{\infty} f_Y(y)\ \mathrm{dy} = 1$.
- **CDF** - $\mathbb{P}[X \le x]$ - $F_X(x) = \sum_{u\le x} f_X(u)$ or $\int_{-\infty}^{y} f_Y(u)\ \mathrm{du}$
- **Quantile** - $q_p = F_Y^{-1}(p)$ for prob p s.t. $\mathbb{P}[Y \le q_p] = F_Y(q_p) = p$
- **Joint pmfs** - $f_{XY}(x,y) = \mathbb{P}[X=x \cap Y=y]$
- **Joint pdfs** - $f_{XY}(x,y) = \mathbb{P}[(X,Y) \in T \subseteq \mathbb{R}^2] = \iint_{(x,y)\in T} f_{XY}(x,y)\ \mathrm{dxdy}$
- **Marginal pmf/pdf** - $f_X(x) = \sum_y f_{XY}(x,y)$ or $\int_{-\infty}^{\infty} f_{XY}(x,y)\ \mathrm{dy}$
- **Conditional pmf/pdf** - $F_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)}$
- **Independence** - $\boldsymbol{X} = (X_1,\ldots,X_n)$ ind. iff $F_{\boldsymbol{X}}(\boldsymbol{x}) = \prod_{i=1}^n f_{X_i}(x_i)$
- X,Y ind. $\implies \mathrm{Cor}(X,Y)=0$, $\mathrm{Cor}(X,Y)=0 \not\Rightarrow X,Y$ ind.

Theorem: Laws Of Large Numbers

Suppose $X_1,X_2,\ldots,X_n \overset{iid}{\sim} F_X$ are s.t. $E[X_i]=\mu$.

- **Weak**: $\overline{X}_n \xrightarrow{p} \mu$ as $n\to\infty$, ie for $\epsilon>0$, $\lim_{n\to\infty}\mathbb{P}[|\overline{X}_n - \mu| < \epsilon] = 1$.
- **Strong:** $\overline{X}_n \xrightarrow{a.s.} \mu$ as $n\to\infty$, ie $\mathbb{P}[\lim_{n\to\infty}\overline{X}_n = \mu] = 1$

Difference: Weak law describes convergence of a sequence of probabilities, strong law describes how a sequence of RVs behave in the limit.

Major Theorem: Central Limit Theorem

Let $X_1,X_2,\ldots,X_n \overset{iid}{\sim} F_X$ s.t. $\mathbb{E}[X_i]=\mu$ and $\mathrm{Var}(X_i)=\sigma^2 \in (0,\infty)$.

Then $\frac{\sqrt{n}(\overline{X}_n-\mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0,1)$, or $\sqrt{n}(\overline{X}_n-\mu) \xrightarrow{d} \mathcal{N}(0,\sigma^2)$,

where $\xrightarrow{d}$ denotes convergence in distribution (convergence of the cdfs).

Definition: Expectation

$\mathbb{E}[X] := \sum_x x f_X(x)$; $\mathbb{E}[Y] := \int_{-\infty}^{\infty} y f_Y(y)\ \mathrm{dy}$.

$\mathbb{E}[g(X)] = \sum_x g(x) f_X(x)$; $\mathbb{E}[g(Y)] = int_{-\infty}^{\infty} g(y) f_Y(y)\ \mathrm{dy}$

- Expectation is linear ($\mathbb{E}[aX+b] = a\mathbb{E}[X]+b$ for $a,b\in\mathbb{R}$).
- Expectation of sum is sum of expectation: $\mathbb{E}[X+Y] = \mathbb{E}[X]+\mathbb{E}[Y]$.
- If X,Y **ind.** then $\mathbb{E}[XY] = \mathbb{E}[X]\cdot\mathbb{E}[Y]$

Definition: Variance

$\mathrm{Var}(X) := \mathbb{E}[(X-\mathbb{E}[X])^2] = \mathbb{E}[X^2] = \mathbb{E}[X]^2 = \mathbb{E}[X(X-\mathbb{E}[X])]$

- $\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$ for $a,b\in\mathbb{R}$.
- $\mathrm{Var}(aX+bY+c) = a^2\mathrm{Var}(X)+b^2\mathrm{Var}(Y)+2ab\mathrm{Cov}(X,Y)$ for $a,b,c\in\mathbb{R}$

Definition: Covariance And Correlation

$\mathrm{Cov}(X,Y) = \mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])] = \mathbb{E}[XY]-\mathbb{E}[X]\mathbb{E}[Y]$

$\mathrm{Cor}(X,Y) = \mathrm{Cov}(X,Y)/\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}$

- $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$
- $\mathrm{Cov}(aX+b,cY+d) = ac\mathrm{Cov}(X,Y)$ for $a,b,c,d\in\mathbb{R}$.
- Correlation is a unitless measurement in $[-1,1]$ with sign indicating direction of lin. dep. and magnitude measuring strength.

Likelihood-Based Inference

The Likelihood Function

Major Definition: Likelihood And Log-Likelihood Function

Consider a random sample $\boldsymbol{Y} = (Y_1,\ldots,Y_n)$ from a distribution with pdf $f_{\boldsymbol{Y}}(\cdot,\theta)$ with $\theta\in\Theta$ unknown. Given a vector $\boldsymbol{y}$, a realisation of $\boldsymbol{Y}$, the **likelihood function** is $L(\theta;\boldsymbol{y}) = f_{\boldsymbol{Y}}(\boldsymbol{y};\theta)$, ie the evaluation of the pdf/pmf at $\boldsymbol{y}$. The **log-likelihood function** $\ell(\theta;\boldsymbol{y})$ is the natural logarithm of the likelihood function.

- Note the likelihood function is **not** a pdf/pmf
- If $Y_1,\ldots,Y_n \overset{iid}{\sim} f_Y(y;\theta)$ then $L(\theta;\boldsymbol{y}) = \prod_{i=1}^n f_Y(y_i;\theta)$ and $\ell(\theta;\boldsymbol{y}) = \sum_{i=1}^n \ln f_Y(y_i;\theta)$.

Major Definition: Maximum Likelihood Estimator (MLE)

For a parameter θ, the MLE of θ is $\hat{\theta} := \mathrm{argmax}_{\theta\in\Theta} L(\theta;\boldsymbol{y}) = \mathrm{argmax}_{\theta\in\Theta}\ell(\theta;\boldsymbol{y})$

- In the univariate case, if L/ℓ are diffable w.r.t. θ, then $\hat{\theta}$ is the solution to $\left.\frac{\partial\ell(\theta;\boldsymbol{y})}{\partial\theta}\right|_{\theta=\hat{\theta}} = 0$.

  Ensure $\hat{\theta}$ corresponds to a maximum! (Check $\ell''(\hat{\theta};\boldsymbol{y})) < 0$.
- If the MLE is not unique for all values of θ the model is non-identifiable

Definition: Identifiability Of A Model

A probability model $\{f(y;\theta) : \theta\in\Theta\}$ is **identifiable** if the mapping $\theta \to f(y;\theta)$ is one-to-one (for any $\theta_1,\theta_2\in\Theta$), if $f(y;\theta_1) = f(y;\theta_2)\forall y$ then $\theta_1 = \theta_2$). Otherwise it is called **non-identifiable**.

- Generally MLEs are not unique in non-identifiable models (ie more than one value maximises L/ℓ)

Example: Checking Model Identifiability

See Example 1.13 in notes - will type up if it seem useful :)

Theorem: Likelihood Principles

Weak: For a model $f(\cdot;\theta)$; two separate realisations $\boldsymbol{x},\boldsymbol{y}$ of rv Y s.t. $L(\theta;\boldsymbol{x}) \propto L(\theta;\boldsymbol{y})$ must lead to the same inferential conclusions (ie same MLE).

Strong: Generalises to case where likelihoods are associated to different models - given $\boldsymbol{x}$ observed from model $f(\cdot;\theta)$ and $\boldsymbol{z}$ from model $g(\cdot;\theta)$ s.t. $L_f(\theta;\boldsymbol{x}) \propto L_g(\theta;\boldsymbol{z})$, they must lead to the same inferential conclusions (same MLE).

Properties Of The Likelihood And MLE

Definition: The Random Perspective

We may treat the lik func, MLE, and their derivatives such as the score function, as random variables dependent on the rv $\boldsymbol{Y}$. We replace $\boldsymbol{y}$ with $\boldsymbol{Y}$ in the definitions, and our new random variables have distributions dependent on the distribution of $\boldsymbol{Y}$.

Major Definition: The Score Function

The **score function** is $U(\theta;\boldsymbol{y}) := \frac{d\ell(\theta;\boldsymbol{y})}{d\theta}$, parametrised by the observations $\boldsymbol{y}$.

- $\hat{\theta}$ can be found by solving $U(\hat{\theta};\boldsymbol{y}) = \left.\frac{d\ell(\theta;\boldsymbol{y})}{d\theta}\right|_{\theta=\hat{\theta}} = 0$
- $\mathbb{E}[U(\theta;Y)] = 0$ for well-behaved pdfs
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